13.4. COSMOGENIC RADIONUCLIDES

13.4.1. Survey

Cosmic irradiation of the atmosphere produces neutrons and protons which react with N_2, O_2, Ar, etc. resulting in the production of radioactive nuclides, some of which are listed in Tables 13.1 and 13.2. These nuclides are produced at constant rates and brought to the earth surface by rain water. Though they are formed in extremely low concentrations, the global inventory is by no means small (§§13.4.2 and 13.4.3). Equilibrium is assumed to be established between the production rate and the mean residence time of these radionuclides in terrestrial reservoirs (the atmosphere, the sea, lakes, soil, plants, etc) leading to constant specific radioactivity of the elements in each reservoir. If a reservoir is closed from the environment, its specific radioactivity decreases. This can be used to determine exposure times of meteorites to cosmic radiation (and the constancy of

Table 13.1 Long-lived cosmogenic radionuclides appearing in meteorites and rain water

Nuclide	Half-life (years)	Decay mode & particle energy (MeV)	Atmospheric production rate (atoms m^{-2} s^{-1})
^{3}H	12.32	β^- 0.0186	2500
^{10}Be	1.52×10^6	β^- 0.555	300
^{14}C	5715	β^- 0.1565	17 000—25 000
^{22}Na	2.605	β^+ 0.545	0.5
^{26}Al	7.1×10^5	β^+ 1.16	1.2
^{32}Si	160	β^- 0.213	1.6
^{35}S	0.239 (87.2 d)	β^- 0.167	14
^{36}Cl	3.01×10^5	β^- 0.709	60
^{39}Ar	268	β^- 0.565	56
^{53}Mn	3.7×10^6	EC (0.596)	
^{81}Kr	2.2×10^5	EC (0.28)	

Values within parenthesis after EC are decay energies.

Table 13.2 Short-lived cosmogenic radionuclides appearing in rain water

	Decay mode	
Nuclide	Half-life	Particle energy (MeV)
^{7}Be	53.28 d	EC (0.862)
^{24}Na	14.96 h	1.389
^{28}Mg	21.0 h	0.459
^{32}P	14.28 d	1.710
^{33}P	25.3 d	0.249
^{39}Cl	55.6 min	1.91

Value within parenthesis after EC is decay energy
Production rates (atoms m^2 s^1): ^{7}Be 81, ^{39}Cl 16

the cosmic radiation field, using ^{81}Kr), dating marine sediments (using ^{10}Be, ^{26}Al), groundwater (^{36}Cl), glacial ice (^{10}Be), dead biological materials (^{14}C), etc. The shorter-lived cosmogenic radionuclides have been used as natural tracers for atmospheric mixing and precipitation processes (e.g. ^{39}Cl or ^{38}S). Only T and ^{14}C are of sufficient importance to deserve further discussion.

13.4.2. Tritium

Satellite measurements have shown that the earth receives some of the tritium ejected from the sun. Much larger amounts are formed in the atmosphere through nuclear reactions; e.g., between fast neutrons and nitrogen atoms

$$\mathrm{n(fast)} + {}^{14}\mathrm{N} \rightarrow {}^{12}\mathrm{C} + {}^{3}\mathrm{H} \tag{13.2}$$

The yield for this reaction is about 2500 atoms tritium per second per square meter of the earth's surface; the global inventory is therefore about 1.3×10^{18} Bq. Tritium has a half-life of 12.33 y, decaying by weak β^- emission to ^{3}He. It is rapidly incorporated in water, entering the global hydrological cycle. The average residence time in the atmosphere is about 2 y which is a small fraction of the half-life, as once the tritiated water reaches the lower troposphere, it rains out in 5–20 days. If we define 1 TU (Tritium Unit) as 1 tritium atom per 10^{18} hydrogen atoms, 1 TU corresponds to 118 Bq/m^3. Before the advent of nuclear energy, surface waters contained 2–8 TU (an average value of 3.5 TU is commonly used). The tritium content in water now commonly is of the order 20–40 TU. Rainwater contains between 4 and 25 TU, lower at the equatorial zone and increasing with latitude.

Tritium is also a product in the nuclear energy cycle, some of which is released to the atmosphere and some to the hydro sphere. The emissions differ between reactor types (usually in the order HWR > PWR > BWR, see Ch. 19 and 20) and is a function of the energy production. Assuming the annual releases to be 40 TBq/GW$_e$ (Giga Watt electricity) from an average power plant and 600 TBq/GW$_e$ from a typical reprocessing plant, the annual global injection of tritium in the environment was estimated to ~10 PBq in 1992. Though this is a small fraction compared to the natural production of ~150 PBq/y, it causes local increases.

The hydrogen bomb tests conducted in the atmosphere during the decade of the 1950's and early 1960's injected large amounts of tritium into the geosphere; 2.6×10^{20} Bq up to the end of the tests in 1962. This considerably exceeds the natural production inventory.

Before 1952 (first hydrogen bomb tests) the tritium content could be used to date water (i.e. determine when it became isolated from contact with the atmosphere). This was very useful e.g. for determining ice ages. However, due to the much larger content of anthropogenic tritium presently, this is no longer a useful technique for such dating.

Tritium in concentrations as low as 1 TU can be measured in low background proportional counters, and, after isotope enrichment (e.g. by electrolysis of alkaline water, by which tritium is enriched in the remainder), down to 0.01 TU. For very low concentrations mass spectrometry is preferred.

13.4.3. 14-Carbon

^{14}C is produced in the atmosphere by a variety of reactions, the most important being between thermalized neutrons from cosmic radiation and nitrogen atoms:

$$n(\text{slow}) + {}^{14}N \rightarrow {}^{14}C + {}^{1}H \tag{13.3}$$

If we compare this reaction with the one forming tritium by the interaction of fast neutrons with ^{14}N we observe that the results are intuitive. The higher the energy of the incoming projectile, the larger the fraction of the compound nucleus is ejected. The production of ^{14}C is approximately 22 000 atoms per s and m^2 of the earth's surface; the global annual production rate is ~1 PBq, and global inventory ~8500 PBq (corresponding to ~75 tons). Of this amount ~140 PBq remain in the atmosphere while the rest is incorporated in terrestrial material. All living material (incl. body tissue) has a ^{14}C concentration of ~227 Bq/kg. The half-life of ^{14}C is 5715 y; it decays by soft β^- emission (E_{max} 158 keV).

^{14}C is also formed by reaction (13.3) in nuclear tests. From these 220 PBq is assumed to have been injected into the atmosphere up to 1990. This ^{14}C comes to equilibrium with other atmospheric carbon (CO_2) in 1–2 years. Figure 13.3 shows the measured atmospheric concentration of ^{14}C as function of time after a nuclear weapons test in the Arctic. Some ^{14}C, about 18 TBq/GW_e per year, is also released from nuclear power plants (mainly from HWR, see Ch. 19–22). The global atmospheric value is <300 TBq/y.

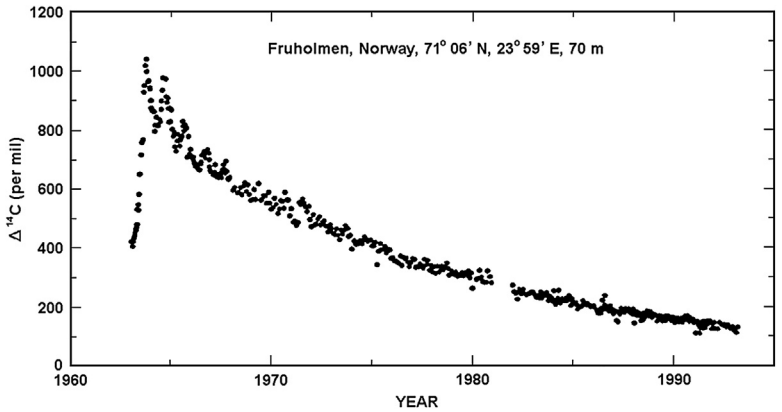


Figure 13.3 Atmospheric concentration of ^{14}C measured at Furuholmen after Soviet nuclear weapons test *(Data from Univ. of Oslo).*

The combustion of fossil fuel adds CO_2, which is almost free of ^{14}C, to the atmosphere, thus reducing the specific activity (the dilution was about 3% for the period 1900–1970). Taking all anthropogenic sources into account, a global average specific activity of modern carbon is now 13.56 ± 0.07 dpm/g C. In §13.8.1 we discuss dating by the "C14-method", for which such a figure is important.

13.5. PRIMORDIAL RADIONUCLIDES

Some radionuclides have halflives so long that some of what was formed in the aftermath of the Big Bang and later supernova explosions still remains on the Earth, and some of their daughters.

13.5.1. Very long-lived nuclides lighter than lead[2]

As the detection technique for radioactivity has been refined, a number of long-lived radionuclides have been discovered in nature. The lightest have been mentioned in §13.4.1. The heavier ones, not belonging to the natural radioactive decay series of uranium and thorium, are listed in Table 13.3. ^{50}V is the nuclide of lowest elemental specific activity (~0.0001 Bq/g) while the highest are ^{87}Rb and ^{187}Re (each ~900 Bq/g). As our ability to make reliable measurements of low activities increases, the number of elements between potassium and lead with radioactive isotopes in nature can be expected to increase.

Table 13.3 Primordial radionuclides for $Z < 82$ (Pb)

Nuclide	Isotopic abundance %	Decay mode and particle energy MeV	Half-life (years)
^{40}K	0.0117	β^- EC 1.31	1.26×10^9
^{50}V	0.250	β^- EC (0.601)	$>1.4 \times 10^{17}$
^{87}Rb	27.83	β^- 0.273	4.88×10^{10}
^{115}In	95.72	β^- 1.0	4.4×10^{14}
^{123}Te	0.905	β^- EC (0.052)	1.3×10^{13}
^{138}La	0.092	β^- EC	1.06×10^{11}
^{144}Nd	23.80	α	2.1×10^{15}
^{147}Sm	15.0	β^- 2.23	1.06×10^{11}
^{148}Sm	11.3	β^- 1.96	7×10^{15}
^{176}Lu	2.59	β^-(1.188)	3.8×10^{10}
^{174}Hf	0.162	α	2×10^{15}
^{187}Re	62.60	β^- 0.0025	4.2×10^{10}
^{190}Pt	0.012	α	6.5×10^{11}

Values within parenthesis refer to decay energies.

[2] The heavier elements are discussed in §13.5.2 and in Ch. 14.

[3] The student can easily trace the decay series in Fig. 13.4 with a transparent marker.